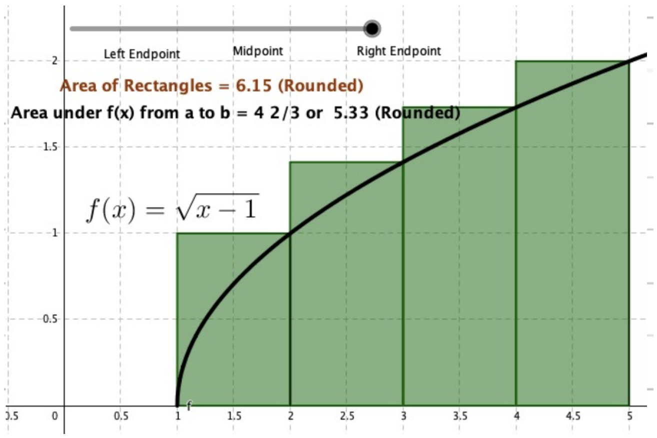
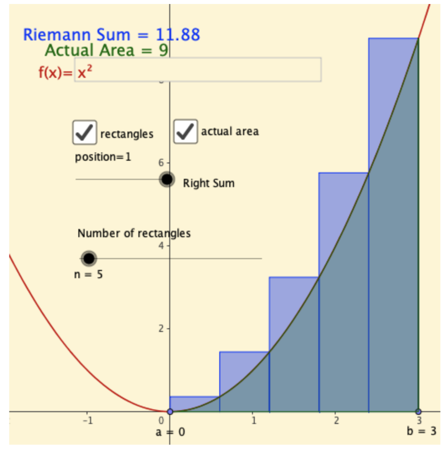
1.1 Approximating Areas

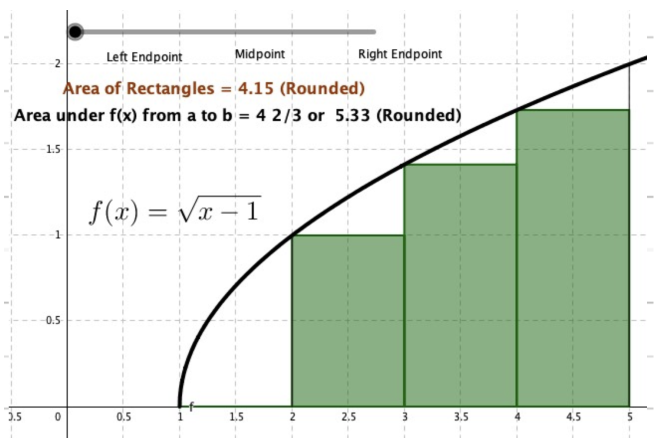
Let's visualize specific types of approximations:

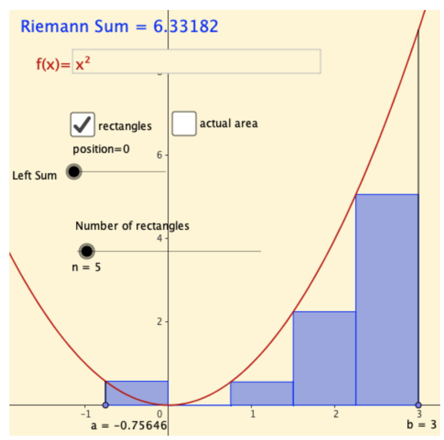
Example 1. Right-endpoint approximation.



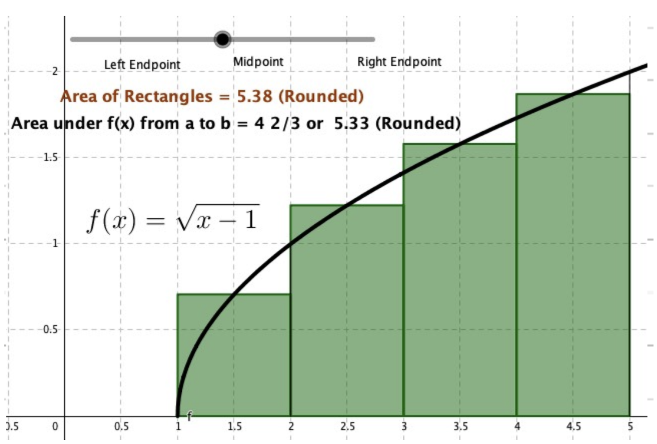


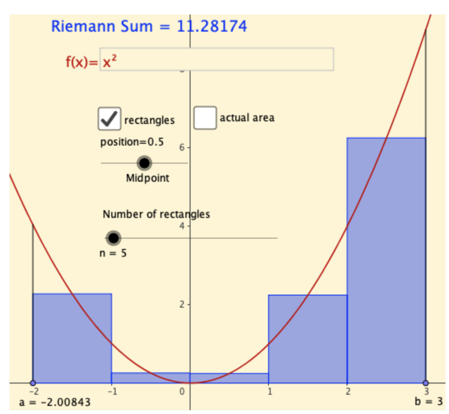
Example 2. Left-endpoint approximation.



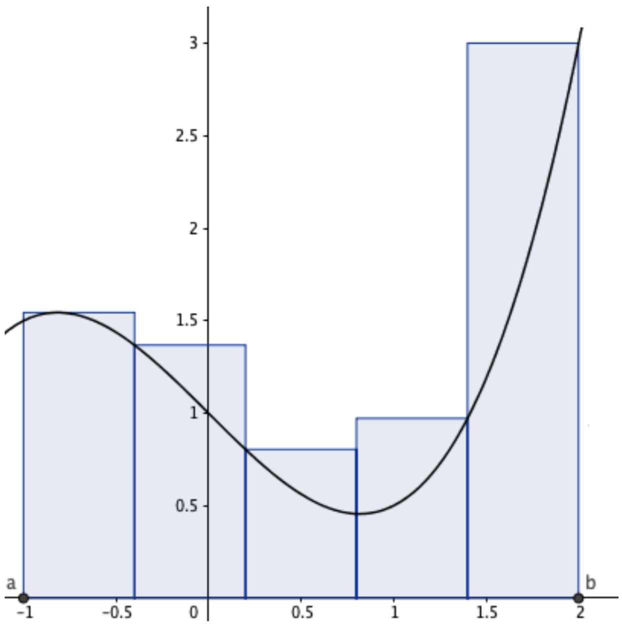


Example 3. Midpoint approximation.

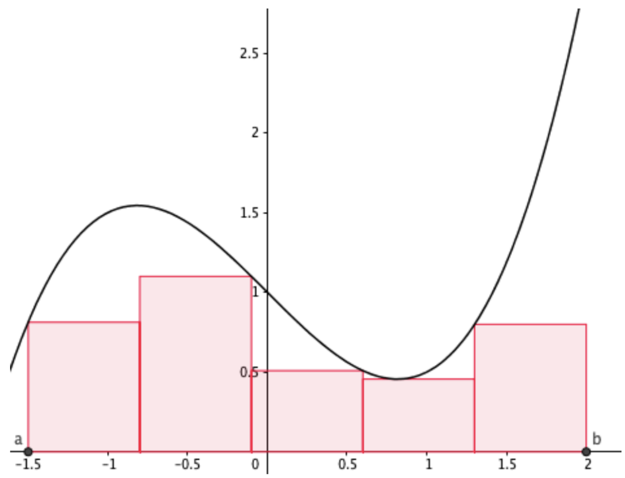




Example 4. **Upper** and **Lower** Sums .



Upper Sum



Lower Sum

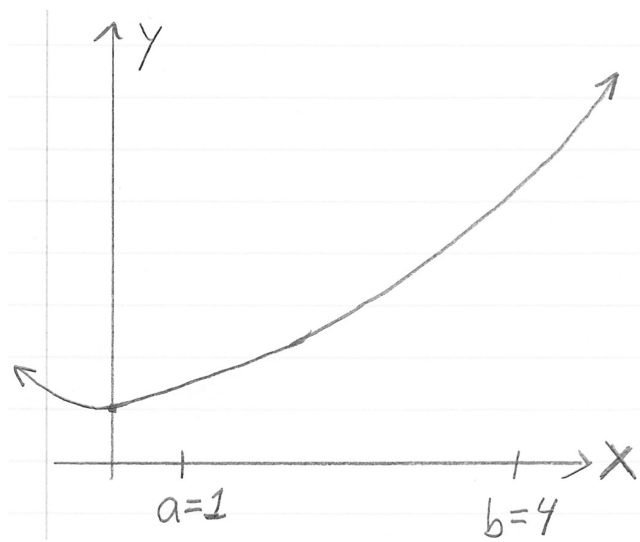
Definition 1. Given n rectangles on the interval $[a, b]$,

- $\Delta x = \dfrac{b-a}{n}$

- $x_i = a + i \cdot \Delta x$,

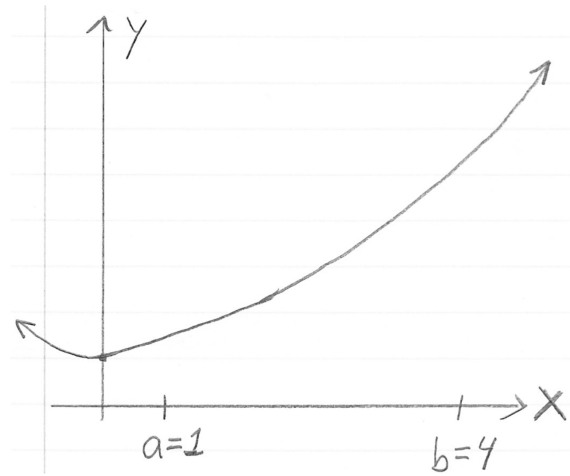
  i = 1, 2, . . . , n

- Notice: $x_0 = a$, $\quad x_n = b$



Example 5. For $y = f(x)$ above, set up **right-endpoint approx.**, $n = 3$ on $[1, 4]$.

Example 6. For $y = f(x)$, set up **right-endpoint approx.**, $n = 4$ on $[1, 4]$.



Example 7. For $g(x) = \cos(\pi x)$, write out and compute the following **right-endpoint approx.** on $[0, 1]$. Can you compute R_5 exactly?

1. $R_4 =$

2. $R_5 =$

Definition 2. Expand the following Riemann sum. Discuss vocabulary...

$$\sum_{i=1}^{4} f(x_i)\Delta x =$$

Example 8. Given $f(x) = \sqrt{x-1}$, expand/write out all the terms.

$$\sum_{k=1}^{3} f\left(1 + k\left(\frac{2}{3}\right)\right) \cdot \left(\frac{2}{3}\right) =$$

Example 9. Write/represent this sum using **sigma notation**.

$$\frac{5}{3}\left[\left(-2 + \frac{5}{3}\right)^2 + \left(-2 + 2\left(\frac{5}{3}\right)\right)^2 + \left(-2 + 3\left(\frac{5}{3}\right)\right)^2\right] =$$

Challenge: In Example 9, identify a candidate for Δx and for $y = f(x)$.

Example 10. Express in sigma notation. Evaluate with RSC: https://www.geogebra.org/m/ezspy4zu

1. R_{10} for $f(x) = \sin(x)$ on $[0, \pi]$

2. R_{20} for $g(x) = \ln(x)$ on $[1, e]$

Riemann sums from tables

Example 11. Use this table of function values (with units) below to write the Riemann sum approximation of the function on $[0, 3]$ with $n = 6$ using the...

x (min)	0	.5	1	1.5	2	2.5	3
$f(x)$ (feet)	1	3	2	5	5.5	6	0

1. The left-endpoint approximation, L_6. Units?

2. The right-endpoint approximation, R_3. Units?

3. The midpoint approximation, M_3. Units?

Example 12. (**Non-Constant Δx_i ?**) Given a table of data with input values non-uniformly spaced, we can "approximate the area" using a right-endpoint approximation with varying (non-constant) Δx_i. Use every entry of the table below to write out the right-endpoint approximation of the function on $[0, 3]$ with $n = 6$.

x (min)	0	0.25	0.75	1	2	2.75	3
$f(x)$ (feet)	1	3	2	5	5.5	6	0

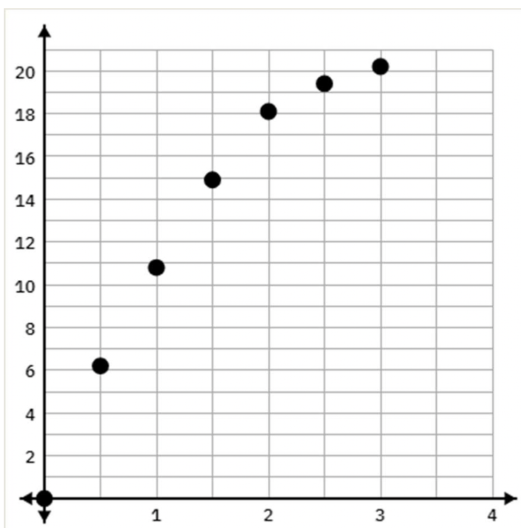
1. Write out the right-endpoint approximation R_6, using every entry.

$$\sum_{i=1}^{6} f(x_i)\Delta x_i =$$

Example 13. The table gives the approximate increase in sea level in inches per 20-years starting in 1870. Estimate the total increase (net change) from 1870 to 1950.

20-years	increase/20-yrs
1870	0.3
1890	1.5
1910	0.2
1930	2.8

Example 14. (**Displacement from a table of values.**) The following table gives the speed of a runner in the first 3 seconds of a race. Using $n = 6$ intervals to estimate the distance travelled during the first 3 seconds:



Time, in seconds	Velocity, in ft/sec
0	0
0.5	6.2
1.0	10.8
1.5	14.9
2.0	18.1
2.5	19.4
3.0	20.2

- **Group 1** - Use the left-endpoint approximation and sketch the rectangles.
- **Group 2** - Use the right-endpoint approximation and sketch the rectangles.

Applications of Riemann sums

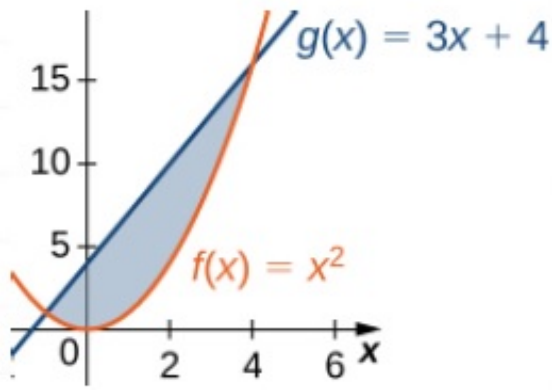
Example 15. Approximate the area between the two graphs with $n = 20$ rectangles:

1. Express as a right Riemann sum using sigma notation, and
2. Calculate the value using the RSC https://www.geogebra.org/m/ezspy4zu

– $g(x) = 3x + 4$ (upper)

– $f(x) = x^2$ (lower)

on the interval $[1, 4]$.



Example 16. Disk method: Area under the curve revolved around the x-axis gives volume.

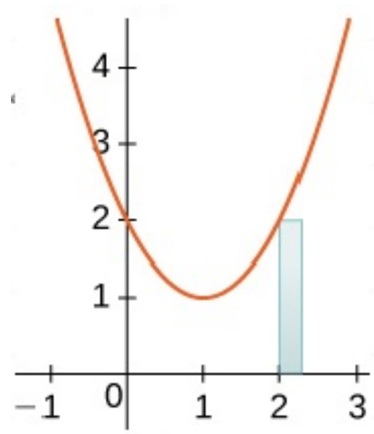
Strategy: A thin "sample rectangle slice" revolved around x-axis gives a cylinder with volume: $V_k = \pi r_k^2 \cdot \Delta x$.

Instructions: Approximate the volume of the solid of revolution around the x-axis with $n = 20$ rectangles by

(1) Expressing it as a right Riemann sum, and

(2) Calculating it with the RSC.

Under $f(x) = x^2 + 1$ on $[-1, 3]$.



1.2 The Definite Integral

Definition 3. Let $f(x)$ be a bounded function on the interval $[a, b]$. We say that f is *integrable* on $[a, b]$ or *the integral of f exists* on $[a, b]$ if the following limit exists:

$$\int_a^b f(x)\, dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*) \cdot \Delta x,$$

for any method of picking x_i^* in the i^{th} interval. **Parts of the integral:** limits of integration, integrand, differential, variable of integration:

Match up symbols: Discuss converting the Riemann sum to a definite integral on $[a, b]$:

$$\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*) \cdot \Delta x$$

Question: Do you know of any integrable functions?

Theorem 1. *If $f(x)$ is a continuous function on the interval $[a, b]$, then $f(x)$ is integrable on $[a, b]$.*

Example 17. Convert to the following:

1. To a definite integral on $[0, 2]$:

$$\lim_{n\to\infty} \sum_{i=1}^{n} 2\ln\left(x_i + 1\right) \cdot \frac{2}{n} =$$

2. To a definite integral on $[1, 4]$:

$$\lim_{n\to\infty} \sum_{i=1}^{n} -3\cos\left(x_i^2\right) \cdot \frac{3}{n} =$$

3. To the limit of a right Riemann sum (right-endpoint approximation):

$$\int_{-1}^{5} x^2\, dx =$$

Geometric Tools...

Since the integral is the area under the curve, we can **use geometric formulae** to compute exact area.

Example 18. Use geometry to evaluate the following (exactly):

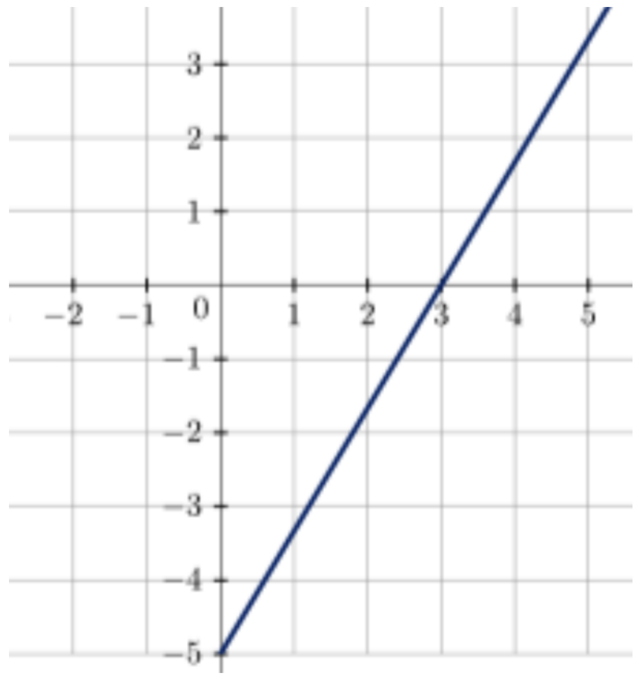
1. $\displaystyle\int_{-3}^{3} \sqrt{9-x^2}\; dx =$

2. $\displaystyle\int_{-3}^{0} 2+\sqrt{9-x^2}\; dx =$

3. $\displaystyle\int_{0}^{6} |x-2|\; dx =$

Example 19. Given is a definite integral and the corresponding graph of the integrand. Identify the function $f(x)$ being graphed. Then use geometry to evaluate the definite integral.

$$\int_0^4 \frac{5}{3}x - 5\ dx$$



Example 20. Evaluate (and visualize).

$$\int_0^2 -\sqrt{4-x^2}\,dx$$

Net Area vs. Total Area

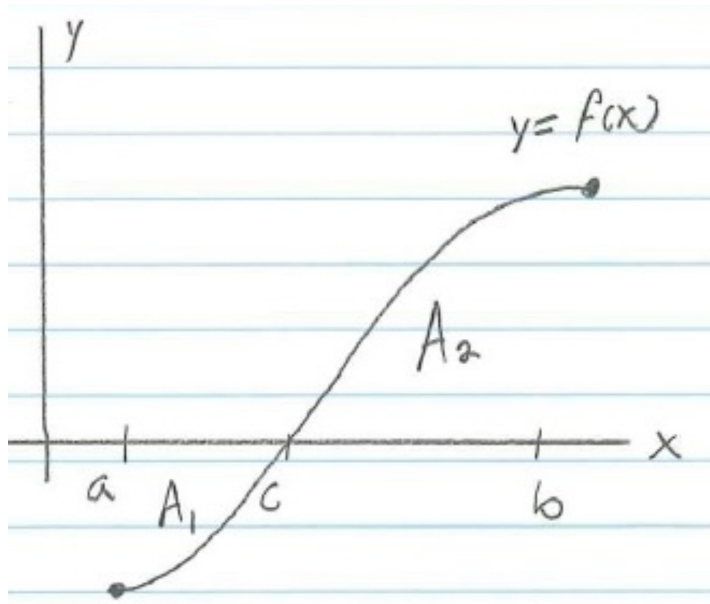
Given is the graph of $y = f(x)$ and (positive) areas A_1 and A_2.

The ***net area*** is

$$\int_a^b f(x)\,dx = -A_1 + A_2.$$

The ***total area*** is

$$\int_a^b |f(x)|\,dx = |A_1| + |A_2|$$



Properties of integrals.

1. $\displaystyle\int_a^a f(x)\,dx = 0$

2. $\displaystyle\int_a^b f(x)\,dx = (-1)\int_b^a f(x)\,dx$

3. $\displaystyle\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

4. $\displaystyle\int_a^b c \cdot f(x) \pm g(x)\,dx =$

Example 21. Express the integral of $f(x) = -2x^{17} + x - 10e^x$ over $[-2, 1]$ as the sum of three integrals.

Example 22. Given the definite integrals $\int_{-1}^{8} h(x)\,dx = 10$, and $\int_{-1}^{5} h(x)\,dx = 2$,

Compute: $\int_{5}^{8} 3 \cdot h(x)\,dx$

Example 23. Given the definite integrals $\int_{-1}^{8} h(x)\,dx = 10$, and $\int_{5}^{-1} h(x)\,dx = 2$,

Compute: $\int_{5}^{8} 3 \cdot h(x)\,dx$

Example 24. Given the definite integrals $\int_{-7}^{2} h(x)\,dx = 3$, and $\int_{-1}^{-7} h(x)\,dx = -6$,

Compute: $\int_{-1}^{2} 2 \cdot h(x)\,dx$

Average value

We can think of the average value of a function $f(x)$ over an interval $[a,b]$ as if the net area between the graph of the function and the x-axis were "flattened" out to make a rectangle of the interval $[a,b]$ with the same area. Then the height of this rectangle is the *average value* of the function (on $[a,b]$).

$$(\text{Ave. height of } f(x)) \cdot (b-a) = \int_a^b f(x)\,dx$$

OR

$$\text{Ave. height of } f(x) \;=\; f_{ave} \;=\; \frac{1}{b-a} \cdot \int_a^b f(x)\,dx.$$

Theorem 2 (Mean-value theorem for integrals)**.** *Let $f(x)$ be a continuous real-valued function on $[a,b]$. Then there is some $c \in [a,b]$ so that*

$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx \qquad \textit{or} \qquad f(c)(b-a) = \int_a^b f(x)\,dx.$$

Example 25. Set up (but do not evaluate) the integral to measure the average value of $f(x) = x - 1$ over the interval $[1,5]$.

Example 26. Find the average value of $f(x) = x^2$ on the interval $[0,3]$. Then find $c \in [0,3]$ satisfying the mean value theorem. (Note. $\int_0^3 x^2\,dx = 9$.)

Example 27. Given f_{ave} of $f(x) = 8 - 2x$ on $[0,4]$ below, find $c \in [0,4]$ satisfying the mean value theorem.

Given: $f_{\text{ave}} = \dfrac{1}{4-0}\displaystyle\int_0^4 8 - 2x\,dx = 4$